

Exploration: Ester and Carbamate Hydrolysis

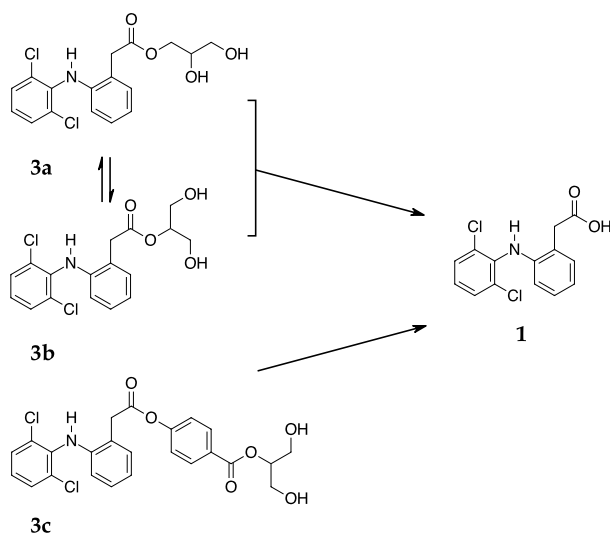
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We are familiar with ester hydrolysis.¹ Does carbamate hydrolysis follow the same mechanism? We will examine the pH-rate profiles of esters and carbamates in two recent publications that explore these topics in the milieu of drug development.

An Example of the pH-Rate Profile for Esters

To demonstrate the pH-rate profile of esters we need some data for rates vs pH to plot. I found such data in a recent contribution examining the hydrolysis of a drug with an ester group within.²

Diclofenac is a powerful NSAID³ that has also been purported to have antibiotic and wound-healing effects. The work focuses on modified diclofenac as models for polymer-bound versions of the drug to study the hydrolysis reaction required for release of the drug from the polymer. Two versions of esterified diclofenac were used, one esterified with glycol, **3a** & **3b**, the other with a phenyl group (that also features a pendant glycol group), **3c**. In water, the ester linkage can be cleaved by hydrolysis. We would expect that the phenyl ester in **3c** would react faster than the alkyl ester in **3a** or **3b**. The rate of hydrolysis of **3a** & **3b** is combined into a single rate due to the interconversion (acyl transfer) reaction.



There are other mechanistic details discussed in the paper, but we will treat this as a simple ester hydrolysis for the purpose of interpreting the data. The pH-rate data is presented in the supplementary material of the paper and reproduced here in table 1 on the next page.

This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were created in ChemDoodle Plots were typeset using Matplotlib and Python tools. Diagrams were created and edited using Affinity Designer.

¹ Read chapter 10.17

² “The Hydrolysis of Diclofenac Esters: Synthetic Prodrug Building Blocks for Biodegradable Drug–Polymer Conjugates,” F. Wang, J. Finnin, et al., *J. Pharm. Sci.*, 2016, 105, 773–785. <https://doi.org/10.1002/jps.24665>

I apologize for choosing this paper. It is badly written with many typos and words switched around. Some paragraphs are unable to be understood as the authors are mixing up their terms. There was clearly no proofreading. This journal should be ashamed to have allowed the paper to be published in this state. The work is good, the writing is awful and the peer review was obviously minimal. It is the editor, however, that should be fired. The editor is responsible for the quality of the journal and they were not doing their job here. Anyway, it contains data for a pH-rate profile for ester hydrolysis done in this century. Beggars can't be choosers.

³ Non-steroidal anti-inflammatory drug

Figure 1: Reaction scheme for the hydrolysis of the esterified Diclofenac, **3a**, **3b** and **3c**.

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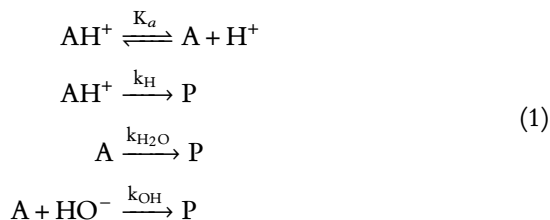
Hydrolysis of 3ab		Hydrolysis of 3c	
pH	$k_{obs_{ab}} / 10^{-5} hr^{-1}$	pH	$k_{obs_c} / 10^{-5} hr^{-1}$
1.00	4270	1.00	2590
1.54	1010	1.90	286
1.90	352	3.10	129
2.80	102	4.04	136
3.22	38.4	4.98	198
3.52	18.9	6.09	1530
4.02	8.46	6.97	12 070
4.45	2.46	7.42	21 460
4.87	3.02	7.96	91 270
5.34	13.3		
5.97	49.1		
6.59	182		
6.93	412		
7.32	1050		
7.82	2880		

Table 1: pH-rate data for the disappearance of **3ab** and **3c**.

←

Data is reproduced from the table found in the supplementary material of the paper.²

The data is plotted in figure 2 on the following page and was fit to one of two models. For the alkyl esters, **3a** & **3b**, the combined rate constant for hydrolysis showed only specific acid and base-catalyzed regions in the plot. The model used is expressed in eq. 2. The phenyl ester, **3c**, showed a high water rate, where there was no rate dependance on acid or base catalysis. The model used is expressed in eq. 3.



$$k_{obs_{3ab}} = \frac{k_H}{K_a} [\text{H}^+] + k_{\text{OH}} K_W \frac{1}{[\text{H}^+]} \tag{2}$$

$$k_{obs_{3c}} = \frac{k_H}{K_a} [\text{H}^+] + k_{\text{H}_2\text{O}} + k_{\text{OH}} K_W \frac{1}{[\text{H}^+]} \tag{3}$$

The best-fit parameters are collected in table 4 on page 9.⁴ We observe that the values in the paper and the values determined using our mathematical are reasonably close to the numbers reported by the authors,⁵ except for the value for k_{OH} in the case of **3c**. It is very different and I can't explain why. Even if I use the value of $K_W = 10^{-14}$ (which is the value at 25 °C) rather than the correct value of $K_W = 10^{-13.62}$, I only get to $k_{\text{OH}} = 1.03 \pm 0.10 \times 10^6$, which is still significantly different than the reported value. If we did use this value of K_W we would also have to apply it to the value of for the k_{OH}

⁴ The authors of the paper fit their data to a series of linear equations in each region of the plots so our values will not be identical.

⁵ Take note that we used a value for the dissociation constant of water at 37 °C of $K_W = 10^{-13.62}$. The value of K_W changes with temperature.

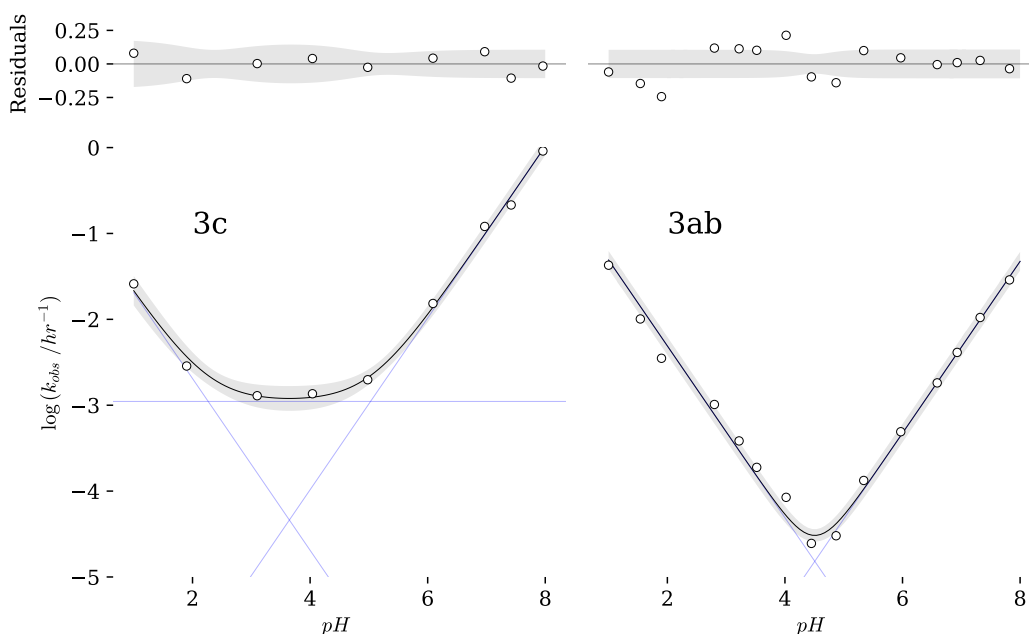


Figure 2: Plots of pH-rate profiles for **3c** and **3ab**. The line is calculates using the best-fit parameters presented in table 4.↑

in the case of **3ab** as well. If we do that, then the value of k_{OH} would be $(4.73 \pm 0.53) \times 10^4$, double the reported value.

The authors do not document their calculations so I cannot say for sure which value of K_W was used and I cannot explain why the value of K_{OH} is so different in that case.

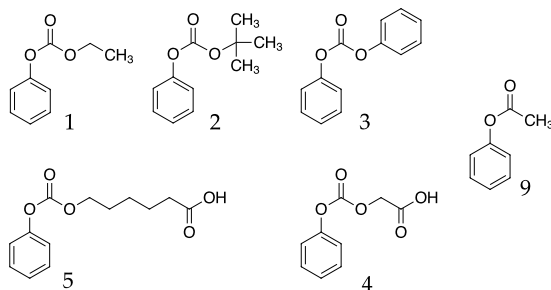
Rate Const.	Hydrolysis of 3ab		Hydrolysis of 3c	
	This work	From paper	This work	From paper
k_H/K_a	$(4.90 \pm 0.54) \times 10^{-1}$	4.50×10^{-1}	$(2.04 \pm 0.34) \times 10^{-1}$	2.04×10^{-1}
k_{H_2O}	—	—	$(1.11 \pm 0.16) \times 10^{-3}$	1.32×10^{-3}
k_{OH}	$(1.97 \pm 0.22) \times 10^4$	2.18×10^4	$(0.432 \pm 0.042) \times 10^6$	1.40×10^6

Table 2: Fit parameters for pH-rate profiles of **3ab** and **3c** compared to literature values.

←

An Example of the pH-Rate Profile for Carbonates

In the search for recent examples of pH-rate profiles for hydrolysis of esters I found a paper detailing the hydrolysis of phenol carbonates as models for modified drugs.⁶ The reactants used in the study are presented in figure 3. I will not be discussing the ester, 9, because that is not a carbonate. Nor will I discuss the *t*-butyl carbonate, 2, because it undergoes hydrolysis in neutral and acidic conditions via a very different mechanism than the other carbonates.⁷



⁶ “Bioreversible Derivatives of Phenol. 2. Reactivity of Carbonate Esters with Fatty Acid-like Structures Towards Hydrolysis in Aqueous Solutions.” J. Ostergaard, C. Larsen, *Molecules*, 2007, 12, 2396–2412. <https://doi.org/10.3390/12102396>

Figure 3: Reactants used in the study of hydrolysis of phenol carbonates.

⁷ The *t*-butyl carbonate, 2, is an example of the famous BOC protecting group. The molecule was easy to obtain from commercial sources; perhaps that is why it was included in this study. However, given the very different chemistry that it undergoes, it was not appropriate to compare it to the others. The ester, 9, was included so that the reader could get a sense of the relative rate of ester and carbonate hydrolysis. But the BOC phenol, 2, is an orange among apples.

⁸ <https://automeris.io> – I used version 4.

The authors did not provide their experimental data in tabular form and present it only as a plot. I digitized the plot using *WebPlotDigitizer*⁸ to obtain the data. As is almost always the case, The authors did not document their data analysis. Their approach is straightforward and I anticipate that we will be able to replicate their results. The data for the hydrolysis of 1, 3, 4, and 5 are collected in table 3.

Hydrolysis of 1		Hydrolysis of 3		Hydrolysis of 4		Hydrolysis of 5	
pH	$\log(k_{obs}/s^{-1})$	pH	$\log(k_{obs}/s^{-1})$	pH	$\log(k_{obs}/s^{-1})$	pH	$\log(k_{obs}/s^{-1})$
0.43	−6.21	0.43	−4.97	0.14	−5.19	0.46	−6.20
0.83	−6.55	0.85	−4.98	0.26	−5.16	0.85	−6.44
1.14	−6.77	1.14	−5.01	0.46	−5.17	1.17	−6.76
2.16	−7.15	2.16	−5.04	0.85	−4.91	2.13	−7.13
3.04	−7.24	3.02	−4.90	1.17	−4.71	3.04	−7.19
4.01	−7.07	4.01	−4.89	1.45	−4.59	4.04	−7.19
5.01	−7.14	5.01	−5.00	1.85	−4.25	4.52	−7.19
6.03	−6.95	6.03	−4.67	2.16	−3.96	5.01	−7.11
7.40	−6.06	7.43	−4.62	2.50	−3.60	6.03	−6.98
8.31	−5.27	9.82	−2.74	3.04	−3.52	6.91	−6.65
9.79	−3.66	11.50	−1.06	4.01	−3.44	8.45	−5.13
11.50	−2.01	12.49	−0.04	5.01	−3.43	9.99	−3.60
12.49	−1.00			6.91	−3.43	12.78	−0.79
				8.37	−3.40	13.17	−0.48
				9.76	−3.26		
				10.78	−2.76		
				10.96	−2.61		
				11.50	−2.13		
				12.49	−1.05		

Table 3: pH-rate data for the hydrolysis of phenol carbonates 1, 3, 4, and 5. Data was digitized from the plot in the paper.⁶ ↑

The data is plotted in figures 5, 6, 7 and 8 on the following pages. I then plotted them together in figure 9 on page 8 for comparison. We see that

there is little difference in the reactivity between 1 and 5. From the point of view of the carbonate, the alkyl group in 1 and the alkyl group that contains a very distant carboxylic acid group in 5 are chemically identical. The diphenyl carbonate is more reactive and shows a high value for the uncatalyzed water rate. The extra phenyl group is able to stabilize charges forming in the transition state better than alkyl groups.

The outlier is the phenyl acetic acid carbonate, 4. We see a region in the pH-rate profile where general base catalysis by the carboxylic acid group is observed.

In figure 4, the schemes for each reaction that contributes to the overall observed rate are presented. The usual reactions that we have seen previously for ester hydrolysis are available here. The protonated carbonate is activated for nucleophilic attack by water in the specific acid mechanism defined by k_H . The neutral carbonate can be attacked by the weak nucleophile water with the rate constant k_{H_2O} and the neutral carbonate can also be attacked by the hydroxide anion, a strong nucleophile, with the rate constant k_{OH} . In the case of 5, we can also have general base catalysis by the deprotonated carboxylic acid that assists water in attacking the carbonate. The pK_a of the carboxylic acid is quite low because of the electron-withdrawing oxygen of the carbonate group. So this means that the general base is available even in slightly acidic water.

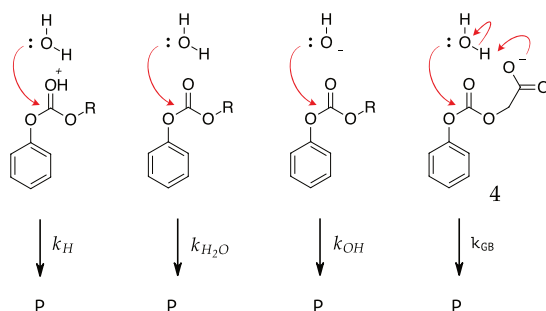


Figure 4: Mechanisms and rate constants for the possible reactions that contribute to the overall observed rate constant, k_{obs} . The general base-catalyzed rate, k_{GB} is only available in the case of 4.

Figures 5 and 6 on the next page present the pH-rate profiles of 1 and 5, respectively. As you can see, they are nearly identical. The pH-rate profile for carbonates 1, 3, and 5 can be fit to the same model described in eq. 3 on page 2.

In figure 7 on page 7 is the pH-rate profile for the diphenyl carbonate, 3. This is a much more reactive carbonate and shows a high rate of hydrolysis via the uncatalyzed mechanism. The curve fit gave a nonsense value for k_H/K_{a1} and was unable to calculate the confidence interval as a result. I removed the k_H/K_{a1} term from the model described in eq. 3 and used the model shown below in eq. 4.

$$k_{obs} = k_{H_2O} + k_{OH}K_W \frac{1}{[H^+]} \quad (4)$$

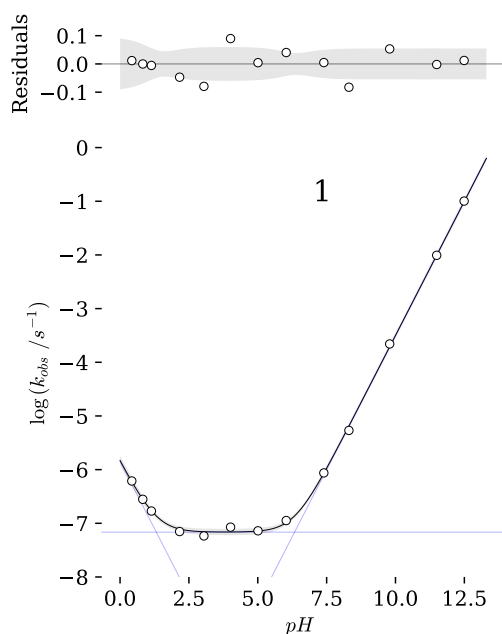


Figure 5: Hydrolysis of 1. The blue lines represent the rates of reaction due to k_H/K_{a1} , k_{H_2O} and k_{OH} . The confidence band is 2σ (95 %.)

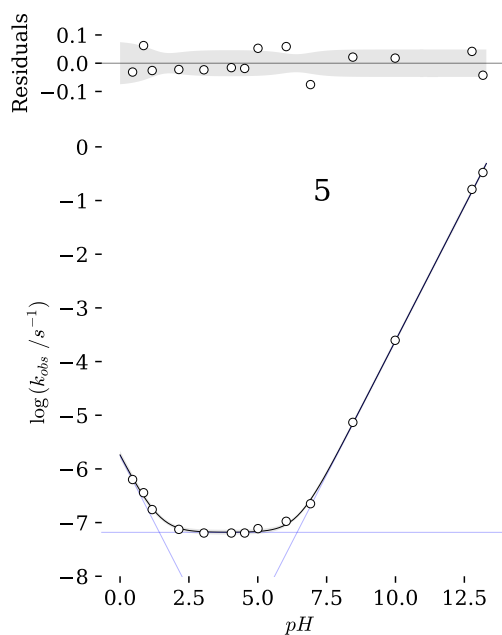


Figure 6: Hydrolysis of 5. The blue lines represent the rates of reaction due to k_H/K_{a1} , k_{H_2O} and k_{OH} . The confidence band is 2σ (95 %.)

In the case of 4, we have the additional rate due to general base catalysis with the rate constant k_{GB} . In this case we can use the model shown in eq. 6.

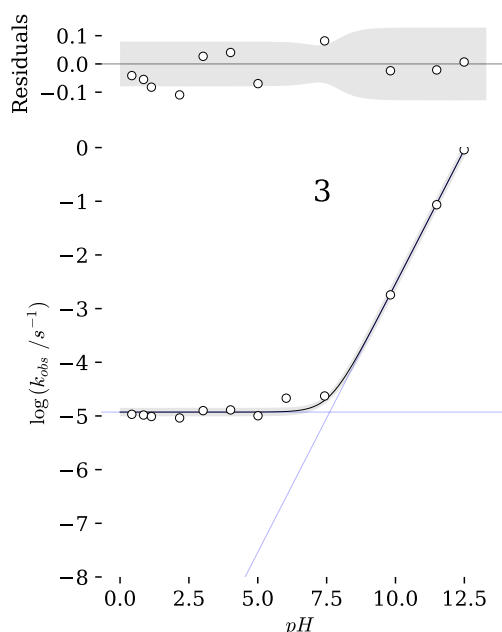
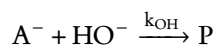
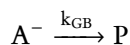
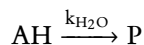
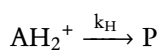
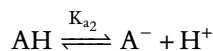
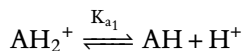


Figure 7: Hydrolysis of **3**. The blue lines represent the rates of reaction due to k_{H_2O} and k_{OH} . The confidence band is 2σ (95 %).



(5)

$$k_{obs} = \frac{k_H}{K_{a1}} [H^+] + k_{H_2O} + \left(k_{GB} + k_{OH} K_W \frac{1}{[H^+]} \right) \frac{K_{a2}}{K_{a2} + [H^+]} \quad (6)$$

On inspection of the data in the case of **4**, it was apparent that there would not be enough information to fit for the rate constants of k_H/K_{a1} and k_{H_2O} .⁹ I modified the model to remove those terms. However the fit was not good. I tried including the k_H/K_{a1} term and the fit was much better. You can try these options by modifying the *Python* notebook that accompanies this document.¹⁰ The final model used is expressed in eq. 7. The plot with the data fitted to the model is presented in figure 8 on the next page.

$$k_{obs} = \frac{k_H}{K_{a1}} [H^+] + \left(k_{GB} + k_{OH} K_W \frac{1}{[H^+]} \right) \frac{K_{a2}}{K_{a2} + [H^+]} \quad (7)$$

⁹ “The word “inspection” implies that I was clever enough to see this issue. In fact it was trial and error. The curve fit fails because of the lack of information to for the value for k_{H_2O} . The rate of general base catalysis is so much faster than the uncatalyzed water rate that the value of k_{H_2O} is unimportant in the case of **4**.

¹⁰ By providing your code you allow others to try other options before inevitably agreeing with you.

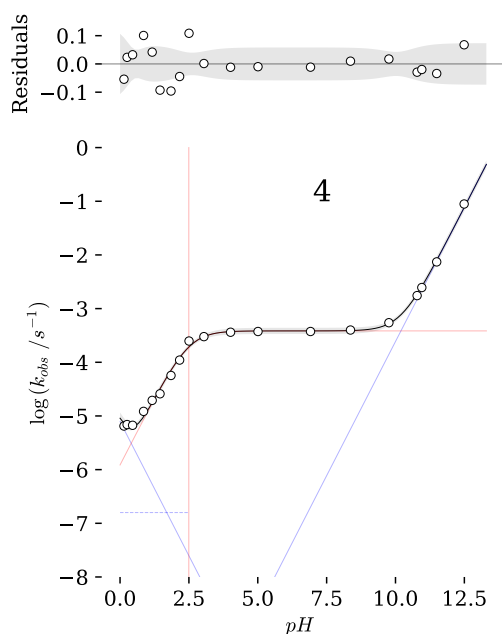


Figure 8: Hydrolysis of 4. The blue lines represent the rates of reaction due to k_H/K_{a1} and k_{OH} .

←
The red lines represent the rates of reaction due to k_{GB} and the value of pK_{a2} . The dotted blue line represents a guestimate of the value of k_{H_2O} for the AH species. The confidence band is 2σ (95 %).

In figure 9 is presented a plot of the lines calculated from the best-fit parameters in each of the previous four plots. How do they compare to you? Do you agree that the left-most detail in the curve fir for 4 is due to the specific acid term, k_H/K_{a1} , and not k_{H_2O} ? Should the acid-catalyzed rate for 3 be more visible, or can you propose a reason why the phenyl ring does not accelerate the specific acid rate as much as the specific base rate, k_{OH} ?

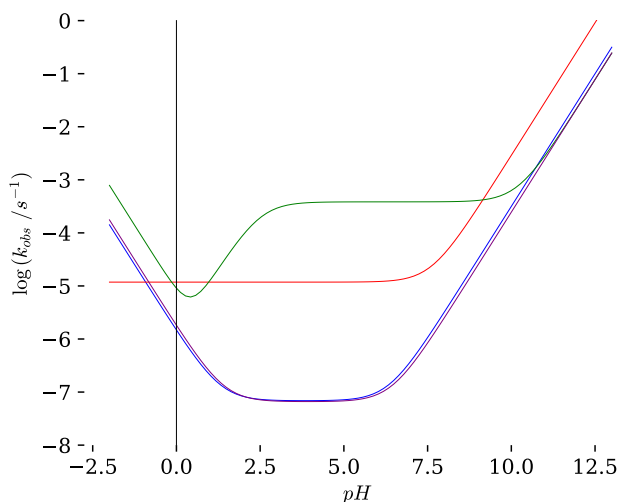


Figure 9: Best fit lines for 1 (blue), 5 (purple), 3 (red) and 4 (green). The vertical black line is the lower limit of the experimental data.

←

Rate Const.	1		5	
	This work	From paper	This work	From paper
k_H/K_{a_1}	$(1.41 \pm 0.14) \times 10^{-6}$	1.4×10^{-6}	$(1.76 \pm 0.14) \times 10^{-6}$	1.7×10^{-6}
k_{H_2O}	$6.85 \pm 0.42 \times 10^{-8}$	6.8×10^{-8}	$(6.58 \pm 0.31) \times 10^{-3}$	6.5×10^{-8}
k_{OH}	1.32 ± 0.07	1.3	1.02 ± 0.05	1.0

Rate Const.	3		4	
	This work	From paper	This work	From paper
k_H/K_{a_1}	—	—	$(3.7 \pm 0.34) \times 10^{-1}$	—
k_{H_2O}	$1.18 \pm 0.09 \times 10^{-5}$	1.2×10^{-4}	—	4.7×10^{-6}
k_{OH}	12.06 ± 1.57	12.6	1.02 ± 0.07	1.0
k_{GB}	—	—	$(3.84 \pm 0.24) \times 10^{-4}$	3.7×10^{-4}
pK_{a_2}	—	—	2.50 ± 0.05	2.6

Table 4: Fit parameters for pH-rate profiles of 1, 3, 4, and 5 compared to literature values.

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